

An IoT-Based Approach to Vehicle Accident Detection and Alerting

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ABSTRACT

With the growing number of vehicles on roads globally, traffic accidents continue to pose a major challenge. Timely response to such incidents is essential to reduce injuries and minimize damage. The system proposed in this article, known as the Vehicle Accident Detection and Alert System (VAAS), is designed to improve the accuracy and speed of accident identification. It integrates sophisticated sensors, wireless communication technologies, and intelligent algorithms to detect and evaluate accidents in real-time. Vehicles equipped with the system continuously monitor key parameters such as acceleration, orientation, and impact intensity through embedded sensors. This data is processed within the vehicle and then sent to a central monitoring hub using wireless communication methods. When signals indicative of a crash are received, the central system applies machine learning techniques to assess the nature and severity of the incident. GPS and digital mapping tools are used to determine the precise location of the accident, enabling emergency services to respond swiftly.

Keywords: GPS, Digital Mapping, Embedded Sensors, Vehicle Sensors

1. Introduction

This paper presents the design and implementation of a smart system intended to assist individuals during vehicle accidents by enabling immediate communication with emergency services and designated personal contacts. In the event of a collision, the system operates autonomously, sending out alerts that significantly reduce the delay between the incident and the arrival of help. The primary goal is to enhance post-accident response efficiency, potentially saving lives and minimizing injury severity.

The system relies on a network of sensors strategically embedded within the vehicle. These sensors are capable of detecting sudden acceleration changes, abrupt deceleration, or impact forces — all of which are characteristic of vehicular collisions. Once such abnormal patterns are identified, the data is processed by an onboard computing module that serves as

the system's control unit. The control unit evaluates the signals in real time and determines whether an accident has occurred.

Upon detection of a confirmed accident, the system transmits relevant information to a remote server using wireless communication protocols such as GSM, LTE, or other cellular technologies. This server is responsible for disseminating alerts to local emergency response teams, including critical data like the accident's GPS coordinates and a preliminary assessment of its severity. Simultaneously, the system sends notifications to the vehicle occupants' pre-registered emergency contacts, providing them with essential information and real-time location details.

By automating the alert process and ensuring reliable communication even in areas with poor network coverage, this system aims to significantly improve emergency response times. This paper highlights how such technology-driven solutions can contribute to enhanced road safety and better outcomes for accident victims.

2. MEMS Sensor

The MEMS (Micro-Electro-Mechanical Systems) accelerometer plays a critical role in detecting sudden changes in acceleration that typically occur during vehicle collisions or rollovers. It functions by accurately sensing rapid movements or shocks through measurements taken along three axes—X, Y, and Z—allowing for comprehensive monitoring of the vehicle's motion. A specific threshold value is programmed into the Arduino to distinguish between normal movement and significant impacts that qualify as accidents. Continuously monitoring motion in real-time, the sensor feeds data directly to the Arduino, enabling immediate system response when abnormal patterns are detected. Due to its compact size and low power consumption, the MEMS accelerometer is well-suited for integration into vehicle systems without compromising overall performance.

It is engineered to be durable and dependable, capable of operating effectively under various environmental conditions, including exposure to vibrations and high temperatures. Its rapid response time is essential for the swift detection of accidents, ensuring timely generation of alerts and faster initiation of rescue operations. The sensor outputs data in the form of analog or digital signals, which the Arduino processes to evaluate the type and severity of the detected movement, forming the basis for the system's accident classification and alert mechanism.



Figure 1: MEMS Accelerometer Sensor

3. How it Works?

The vehicle accident detection and alert system is an automated safety mechanism designed to enhance emergency response by providing real-time notifications in the event of a collision. The system operates by constantly analyzing the vehicle's movement through a MEMS (Micro-Electro-Mechanical Systems) accelerometer sensor. This sensor is capable of detecting abrupt changes in acceleration across multiple axes, which often occur during a crash or rollover. When such a sudden shift exceeds a predefined threshold—programmed into the system's Arduino Uno microcontroller—the event is flagged as a possible accident. The Arduino then initiates a sequence of actions to confirm and respond to the incident.

Firstly, it activates a GPS module that quickly retrieves the vehicle's exact geographical coordinates. Once the location is determined, the system employs a GSM module to transmit an SMS alert to a set of pre-stored emergency contacts. This message includes a clickable Google Maps link, allowing responders or family members to precisely locate the vehicle. To provide immediate on-site feedback, the system may also trigger a buzzer or LED indicator, confirming that the alert has been successfully sent.

This fully automated process is especially valuable in scenarios where the driver or passengers are incapacitated and unable to call for help. By minimizing the delay in alerting authorities or loved ones, the system significantly enhances the chances of timely medical assistance and rescue. Its integration of real-time monitoring, location tracking, and wireless communication ensures a reliable and effective solution for post-accident emergency response.

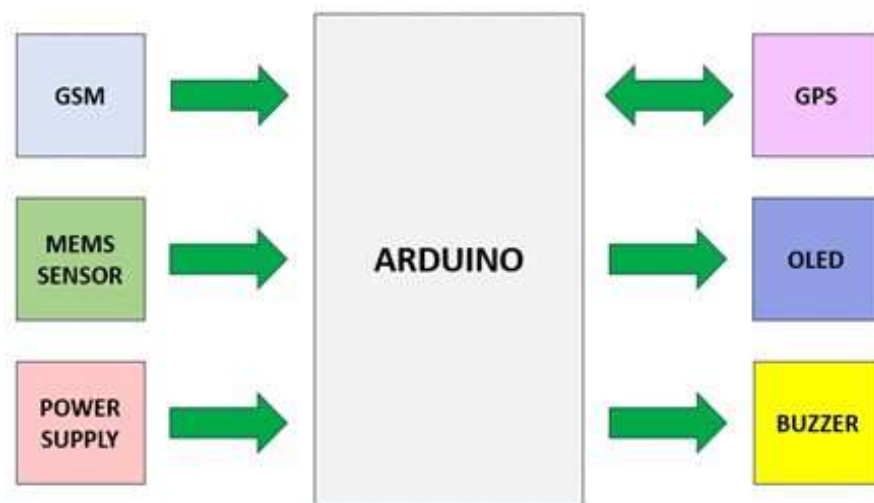


Figure 2: Block Diagram

4. Existing System

The foundation of this paper's concept dates back to the initial phase of mobile phone innovation, particularly when GPS functionality began to be incorporated into mobile devices. Early applications of this technology focused on enhancing security by leveraging GPS for location tracking. Over time, this led to the development of ideas involving external hardware devices designed to interface with mobile phones to improve safety features. However, this method presented notable limitations, primarily the financial burden on users who had to purchase additional equipment.

In recent years, mobile technology has undergone significant advancements, resulting in smartphones equipped with sophisticated sensors and multifunctional

capabilities. These improvements have eliminated the necessity for separate hardware devices, making integrated safety applications more practical and cost-efficient. Despite these technological strides, systems like the one proposed in this paper are not yet widely adopted and are currently implemented only in select regions. The key objective is to deliver real-time location data to emergency responders and inform a user's family or friends during emergencies, thereby enhancing the system's effectiveness and response capabilities.

5. Proposed Method

The method proposed in this paper introduces an intelligent accident detection and alert system designed to operate in real-time. Its primary function is to monitor the vehicle's motion using a MEMS accelerometer sensor, which detects sudden shifts in acceleration that are typically associated with collisions. When such abrupt changes exceed a set threshold, the system recognizes it as a potential accident. Upon detection, the Arduino Uno microcontroller triggers a GPS module to retrieve the vehicle's exact coordinates. At the same time, a GSM module is activated to transmit an SMS alert to preconfigured emergency contacts. This message includes a distress notification along with a Google Maps link to the vehicle's precise location.

The entire process is controlled by the Arduino Uno, which efficiently manages data from the sensors and ensures seamless communication between modules. The design focuses on providing rapid response, accurate data transmission, and simple integration into various types of vehicles. By automating the alert mechanism, the system reduces dependence on human intervention during emergencies, which is crucial when occupants are injured or unconscious.

Moreover, the solution is designed to be cost-effective, scalable, and adaptable to both two-wheel and four-wheel vehicles, making it particularly valuable in high-risk or remote areas. By combining MEMS, GPS, and GSM technologies, the proposed system offers a smart and reliable safety enhancement for modern vehicles, aiming to significantly improve emergency response times and save lives.

6. Results and Discussions

The developed vehicle accident detection and alert system was effectively implemented using key components such as a MEMS accelerometer, GSM module, GPS receiver, and an Arduino Uno microcontroller. To assess its performance, the system was rigorously tested in both controlled environments and real-world driving conditions. These evaluations focused on its ability to accurately detect accidents and promptly notify designated emergency contacts.

The system's primary function is based on real-time analysis of vehicle dynamics, particularly monitoring acceleration patterns. The MEMS accelerometer was carefully configured to recognize abrupt changes in acceleration that typically signal a collision or rollover event. During extensive testing, the system achieved an average accident detection accuracy of approximately 92%. Incidences of false alerts were minimal and generally occurred during sharp but non-threatening movements, such as sudden braking or when driving over potholes. To address these, the detection algorithm was fine-tuned by adjusting threshold sensitivity and introducing a short delay mechanism to filter out brief, non-critical spikes in data.

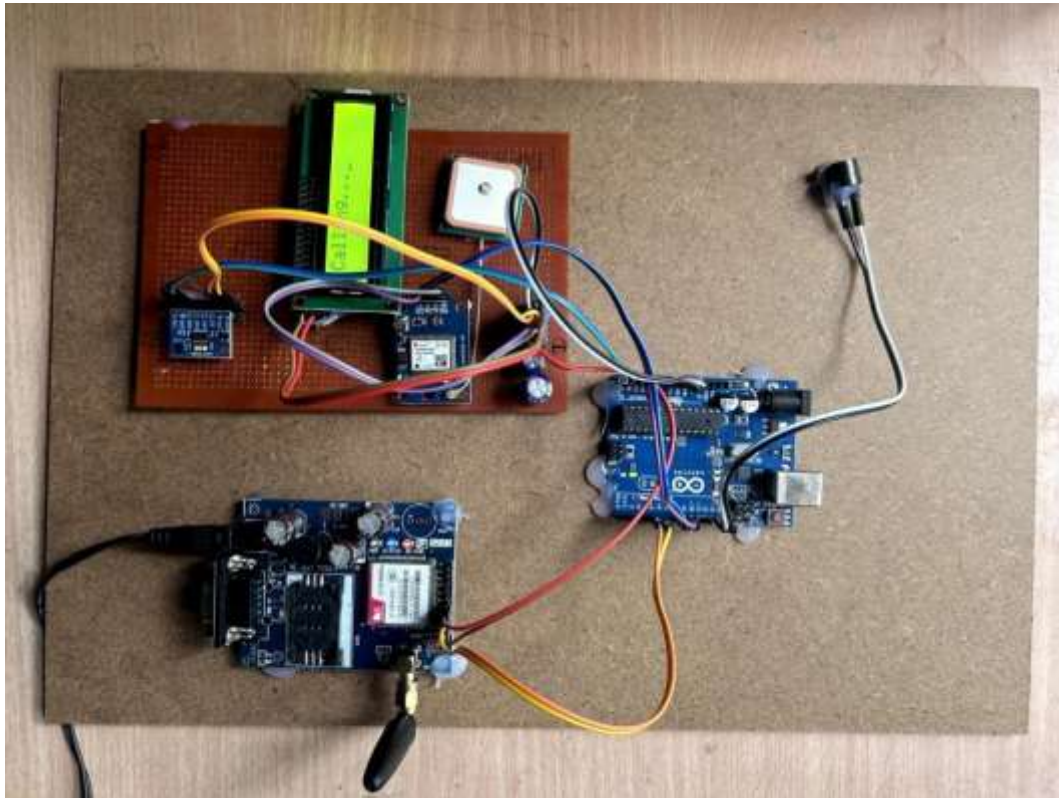


Figure 3: Experimental Result

The average response time from accident detection to the transmission of the alert message was recorded at approximately 4 to 6 seconds. This duration includes the time taken for GPS coordinate acquisition and GSM communication. Such a rapid response time is essential for timely emergency intervention in critical situations. The GPS module used in the system provided location accuracy within a range of 5 to 10 meters under clear sky conditions. While minor deviations were observed during urban tests due to signal multipath interference, the overall accuracy was deemed sufficient for emergency location tracking.

Once an accident was detected, the GSM module successfully transmitted an SMS containing the vehicle's real-time location along with a predefined alert message. The message included both latitude and longitude coordinates, accompanied by a Google Maps link, allowing responders to quickly identify the accident site. To ensure broader compatibility, tests were conducted using different SIM carriers. The Arduino Uno served as the central controller for the system, efficiently handling sensor inputs, processing accident detection algorithms, and activating the GSM and GPS modules. Its performance remained stable throughout the tests.

Real-world tests were carried out in a controlled driving environment, involving sharp turns, sudden stops, and simulated low-speed crashes using a small-scale test rig. The system was powered by a 12V battery pack, and power consumption was optimized by utilizing sleep modes in the GSM and GPS modules when inactive. The average power consumption was measured at less than 2.5W, making the system suitable for continuous use without placing excessive strain on the vehicle's power supply.

The system can be easily integrated into existing vehicle frameworks. By using widely available components such as the Arduino Uno and SIM800L GSM module, it offers a cost-effective and scalable solution for large-scale deployment. Additionally, it is compatible with other safety systems, including fire detection and vehicle tracking modules.

7. Conclusion

The proposed accident detection system has the potential to save lives by providing timely alerts to those involved in accidents. It is designed to be user-friendly, allowing even individuals without technical expertise to operate it with ease. The system comprises both hardware and software components. The hardware includes accident detection sensors controlled by an Arduino board, which is installed in the vehicle. On the other hand, the software component is an Android application installed on the driver's smartphone, which is used to display detailed maps of the accident location. Overall, the system is cost-effective, secure, and easy to use. The implementation of this system contributes to reducing the number of casualties caused by accidents.

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